INCH-POUND
MIL-DTL-24643/15G
27 November 2023
SUPERSEDING
MIL-DTL-24643/15F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSDSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Thermoset insulation, extruded or taped (see table I for thickness).
Third	-	An optional glass braid.
Fourth	-	An optional covering (unless required by 3.3.6 of MIL-DTL-24643).
Fifth	-	The two conductors shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. Standard identification code applied by method 1 or 3, or letter identification code applied by method 5. If identification method 1 is used, outer surface to be printed shall appear black.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/15	Type and size	Conductor size		Cable overall diameter max. (inches)	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	AWG/MCM						
-01UN	LSDSGU-3	--	16 (class B)	0.391	0.018	0.030	4.3	500	N/A
-02UN	LSDSGU-4	--	14 (class B)	0.427	0.018	0.030	2.68	500	N/A
-03UN	LSDSGU-9	--	10 (class B)	0.544	0.018	0.040	1.08	500	N/A
-04UN	LSDSGU-14	--	9 (class B)	0.670	0.018	0.040	0.859	500	100
-05UN	LSDSGU-23	--	7 (class B)	0.781	0.018	0.050	0.543	500	130
-06UN	LSDSGU-50	--	3 (class C)	0.911	0.035	0.050	0.210	200	215
-07UN	LSDSGU-75	--	1 (class C)	1.074	0.035	0.050	0.134	175	285
-08UN	LSDSGU-100	--	0 (class D)	1.167	0.035	0.050	0.106	160	335
-09UN	LSDSGU-200	--	0000 (class D)	1.583	0.050	0.060	0.053	125	525
-10UN	LSDSGU-300	--	300 MCM (class D)	1.841	0.050	0.075	0.0377	110	700
-11UN	LSDSGU-400	400 (127)	--	2.069	0.050	0.075	0.0273	100	835

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	
Sizes 3 through 9	3,000
Sizes 14 and larger	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (scrapes, min.) (sizes 40 and smaller) (extruded insulation only)	250
Crack resistance (sizes 3 through 100) (extruded insulation only)	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (sizes 14 and larger) (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSDSG	3	2,500
LSDSGU	4 through 23	2,000
LSDSGU	50	1,500
LSDSGU	75 and 100	1,000
LSDSGU	200 through 400	800

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-031)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.